

PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
IRINGA REGION
MIDTERM EXAMINATION
FORM ONE
MATHEMATICS

1.(a) what is mathematics?

(b) Choose any 4 subject of your choice other than mathematics and explain with vivid example how each of them is connected to mathematics.

2.(a) converting recurring decimal into fraction $0.8333333\ldots$

(b) If Z is an integer greater or equal to -4 but less than 3 represent on a number line.

3.(a) An electrician used 12,789 m, 13,953 m and 9,374m lengths of wire in 3 different months respectively.

(i) Find the average amount of wire used to nearest hundreds for the 3 different months.

(ii) What is the difference between the actual amount and estimate amount.

(b) There are 64 pipes each with a length 4.22 m arranged horizontally end to end. Estimate the lengths of all the pipes.

4. (a) Jason has 500, 200 and 100 Tanzania shillings coins. The ratio of total amount of the coins 2:3:5 and the total amount is 780,000 Tanzania shillings. Find the number of coins of each kind.

(b) Find A:B:C when A:B = 3:5 and A:C = 6:7.

5. Find the solution of the following simultaneous equations.

$$\begin{cases} \frac{5}{x} - \frac{1}{2y} = \frac{1}{2} \\ \frac{2}{y} + \frac{15}{x} = 5 \end{cases}$$

(a) By use elimination method.

(b) By use substitution method.

6. (a) The slope of the line joining the points (1,p) and (4,2p) is $\frac{1}{3}$. Find the value of p.

(b) A straight line has a gradient of $\frac{3}{4}$ and passes through the point (-1,2). Find

(i) The equation in form $y=mx + c$

(ii) The x and y – intercept.